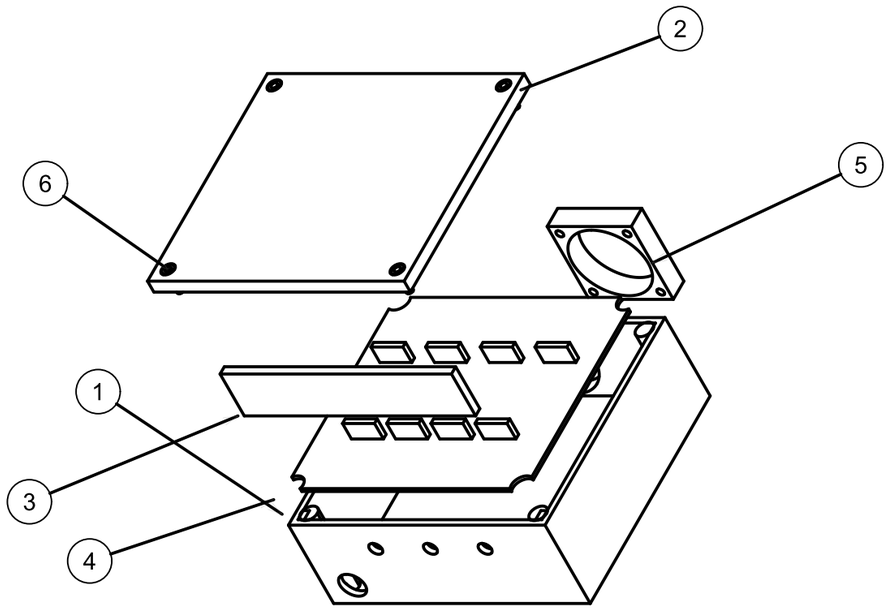
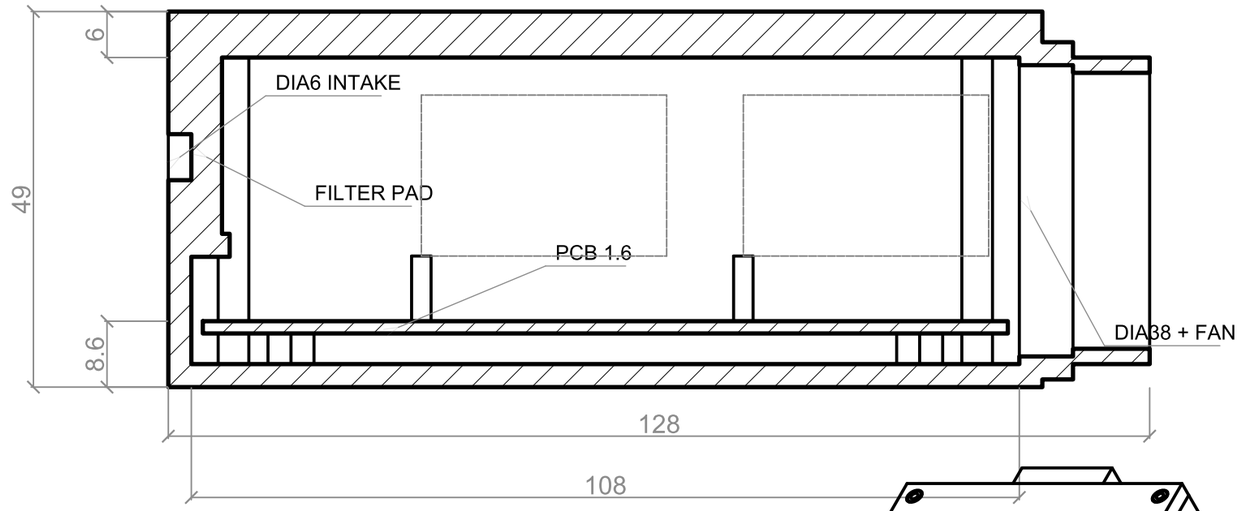
ASSEMBLY – OpenSmell v1.1 sensing platform

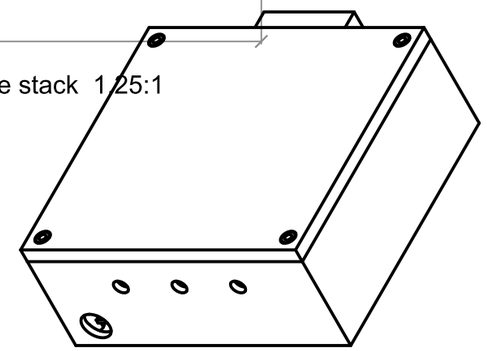
four screws out, lid off, modules in, lid back. That is the entire service procedure.



EXPLODED – assembly order (NTS)



SECTION B-B – x=61, the whole stack 1.25:1



ISO – as built (NTS)

BILL OF MATERIALS

ITEM	PART / DESCRIPTION	QTY	MATERIAL / SOURCE	TYPE
1	Shell – open tray 122 × 114 × 43	1	PETG FDM (or AL 6061-T6)	make
2	Lid – flat plate 122 × 114 × 6	1	PETG FDM, same spool	make
3	Filter pad 110 × 4 × 23	1	non-woven or PTFE felt	buy
4	Main PCB 114 × 105, 8 sockets	1	FR4 1.6 mm + build (OS-003)	make
5	Fan DIA40 × t10, 5 V	1	any 40 mm axial	buy
6	Screw M3 × 12, socket head	4	A2 stainless	buy

NOTES

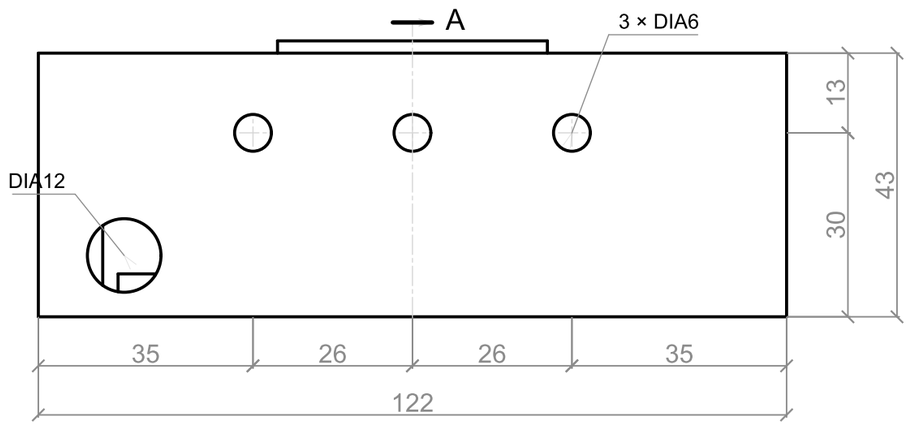
1. ENVELOPE 122 × 114 × 49, FAN 10 PROUD OF THE BACK.
2. 8 SOCKETS: 1×6 KEYED 2.54 mm, ANALOG ROW DOWNSTREAM.
3. LID SCREWS: M3 × 12 INTO HEAT-SET INSERTS, 4 PLACES.
4. FILTER DROPS INTO THE FRONT DRAWER; NOTHING ELSE IS A CONSUMABLE.
5. NO SILICONE, PVC, PAPER OR FOAM WHERE THE SAMPLE TOUCHES.
6. BALLOON NUMBERS MATCH THE BOM; THE EXPLODED VIEW IS TRUE HLR.
7. DASHED = OSM-1 MODULE (USER-SUPPLIED); 4.9 mm AIR ABOVE ITS FACE.

GENERAL TOLERANCE ISO 2768-mK · PROJECT OpenSmell v1.1 REV M
NOTES: all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

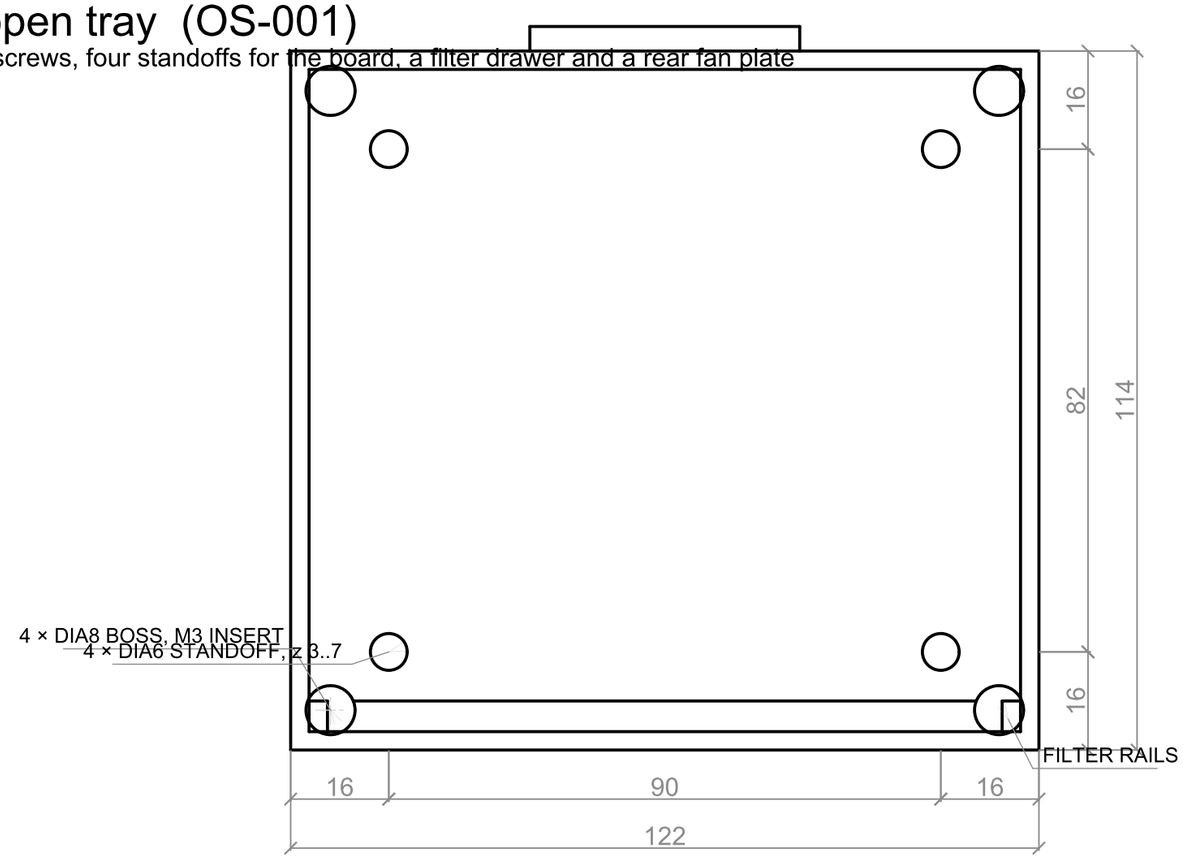
ASSEMBLY			SCALE AS NOTED
DRAWING No. OS-000	REV M	SHEET 1/3	SIZE A3 UNITS mm
MATERIAL see BOM FINISH see BOM			FIRST ANGLE (ISO 128)
DRAWN OpenSmell	CHECK –	DATE 2026-10-21	

SHELL – open tray (OS-001)

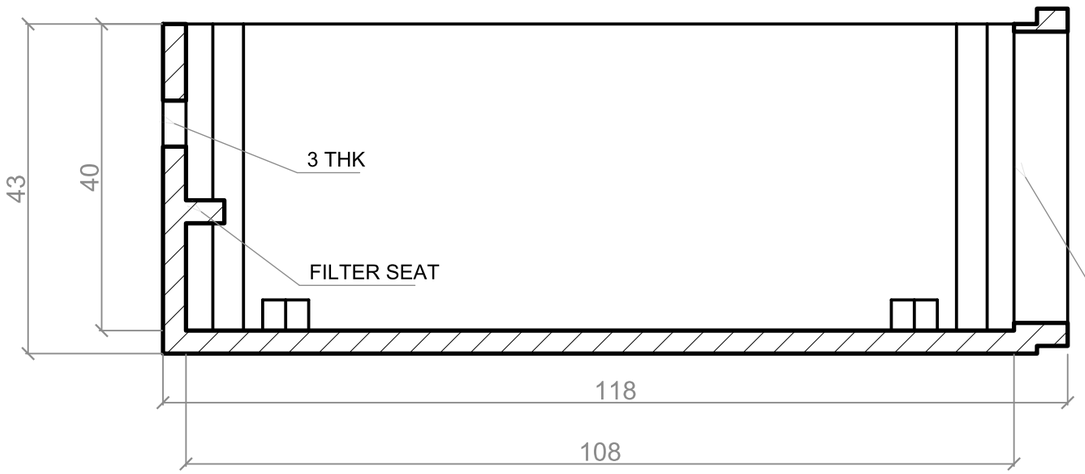
one printed part: floor, four walls, four bosses for the lid screws, four standoffs for the board, a filter drawer and a rear fan plate



FRONT – intake face 1:1



PLAN – open top 1:1



SECTION A-A – x=61 1.25:1

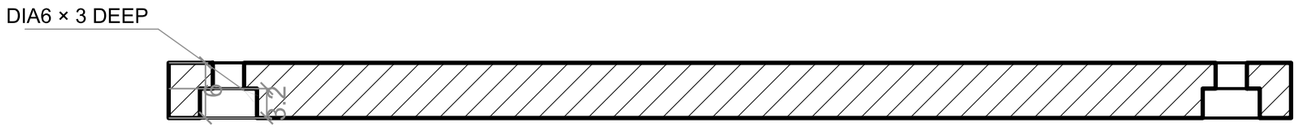
NOTES
1. MATERIAL: PETG FDM, 0.2 mm LAYER, 4 PERIMETERS; SEAL WETTED FACES.
2. WALL 3, FLOOR 3, LID 6 (OS-002); CAVITY 116 × 108 × 40.
3. BOARD SITS ON 4 STANDOFFS (DIA6, z=3..7), TOP FACE = CHAMBER FLOOR.
4. LID: 4 × M3 × 12 INTO HEAT-SET INSERTS IN DIA8 CORNER BOSSES; LID ADDS 6 mm.
5. FILTER PAD 110 × 4 × 23 DROPS IN FROM THE TOP, HELD BY THE LID.
6. NO SILICONE, PVC, PAPER OR FOAM IN THE GAS PATH. DIMS IN mm.

GENERAL TOLERANCE ISO 2768-mK · PROJECT OpenSmell v1.1 REV M
NOTES: all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

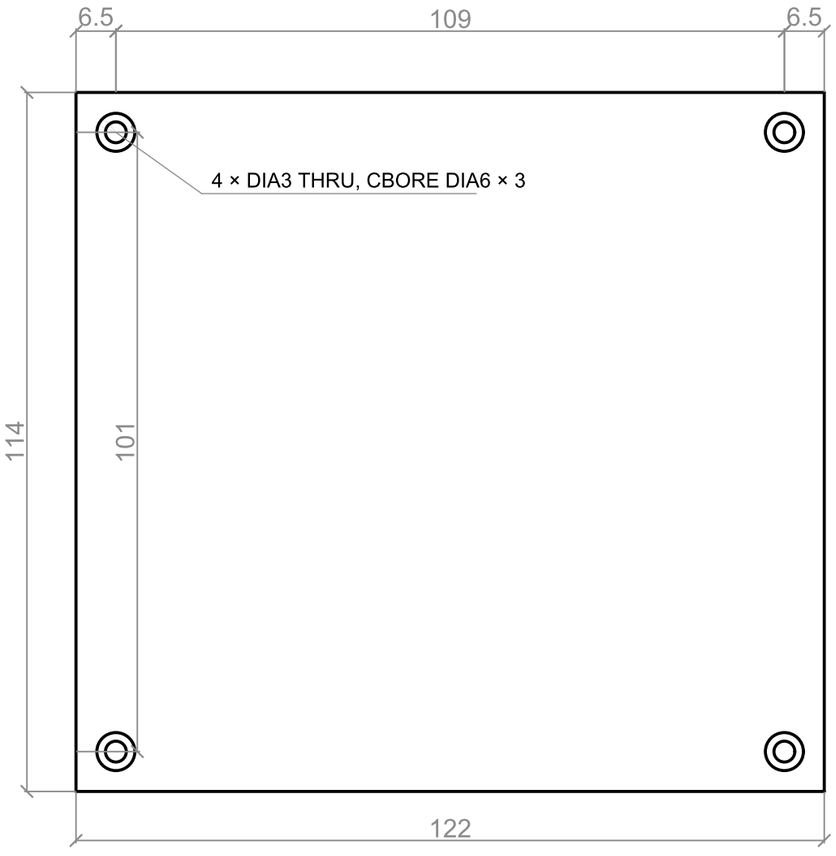
SHELL – open tray			SCALE AS NOTED
DRAWING No. OS-001	REV M	SHEET 2/3	SIZE A3 UNITS mm
MATERIAL PETG, FDM 0.2 mm, 4 perimeters, vapor-sealed inner cavity FINISH as-printed, de-flash; wetted faces seamed/sealed			FIRST ANGLE (ISO 128)
DRAWN OpenSmell	CHECK –	DATE 2026-10-21	

LID (OS-002)

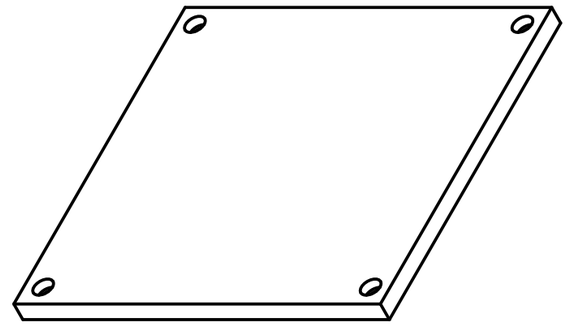
a flat plate with four counterbored holes: it closes the chamber, holds the filter down and clamps nothing else



SECTION C-C – y=6.5, through the counterbores 1.5:1



PLAN – hole pattern 1:1



ISO – lid as printed (NTS)

NOTES

1. MATERIAL: PETG FDM, 0.2 mm LAYER, 4 PERIMETERS (SAME SPOOL AS SHELL).
2. PLATE 6 THICK; COUNTERBORE 6 × 3 FOR M3 SOCKET HEAD.
3. MATING FACE FLAT WITHIN 0.2; NO GASKET - THE FAN RUNS AT NEGATIVE PRESSURE.
4. THE UNDERSIDE HOLDS THE FILTER PAD DOWN: NO ADHESIVE, NO FASTENER.

GENERAL TOLERANCE ISO 2768-mK · PROJECT OpenSmell v1.1 REV M
NOTES: all dims mm; deburr/de-flash; seal wetted faces; print thin parts 0.2 mm layer

LID – flat plate			SCALE AS NOTED
DRAWING No. OS-002	REV M	SHEET 3/3	SIZE A3 UNITS mm
MATERIAL PETG, FDM 0.2 mm, 4 perimeters (same spool as the shell) FINISH as-printed; mating face flat within 0.2			FIRST ANGLE (ISO 128)
DRAWN OpenSmell	CHECK –	DATE 2026-10-21	